

Kinetic Modeling of Cell Signaling

Module 1: Reaction networks as dynamical systems

Jim Faeder

MSCBIO 2025 · Introduction to Bioinformatics Programming in Python

Fall 2026

Resources

- ▶ **BioNetGen:** software, documentation, and tutorials: bionetgen.org
- ▶ **VS Code extension:** search the Extensions panel for *BioNetGen*
- ▶ **bngsim:** fast ODE and stochastic simulation of BioNetGen models from Python: bngsim.readthedocs.io
Install with `pip install bngsim bionetgen`
- ▶ **Course repository:** github.com/jrfaeder/RBM_Intro_BioNetGen
 - `models/` BNGL models used in the slides
 - `notebooks/` Jupyter notebooks for simulation and fitting
 - `Slides/pdf/` slides and handouts



Course repository
models, notebooks, slides

Goals for this module

By the end of this module you should be able to

1. describe a biochemical process as a set of **species** and **reactions**
2. write the rate of each reaction using **mass-action kinetics**
3. turn a reaction network into a system of **ordinary differential equations**
4. encode the same network in the **BioNetGen language** (BNGL), simulate it, and interpret the results
5. explain how a **negative feedback** loop shapes the dynamics of a signaling response

Part 1

Why model reaction networks?

Reaction networks describe how molecule populations change

A **reaction network model** consists of

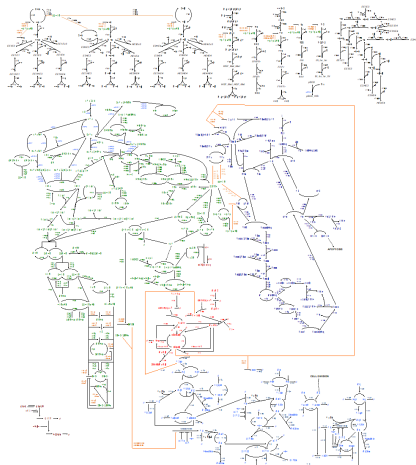
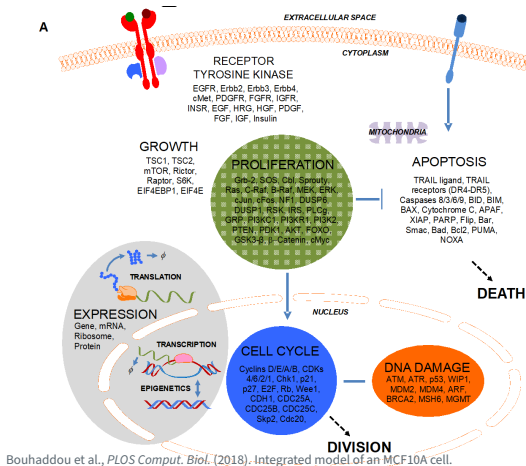
- ▶ **species:** the molecules or complexes we track
- ▶ **reactions:** how species are converted into one another
- ▶ **rate laws** and **rate constants:** how fast each reaction occurs
- ▶ **initial conditions:** the amount of each species at $t = 0$

The same framework describes atmospheric chemistry, epidemics, ecology, and the signaling and gene-regulatory networks that control cell behavior.

What a model gives us

- ▶ A precise statement of a mechanism
- ▶ Predictions of dynamics we cannot easily measure
- ▶ A way to test whether a mechanism can explain data
- ▶ Hypotheses for the next experiment

Large models are built from simple reaction steps

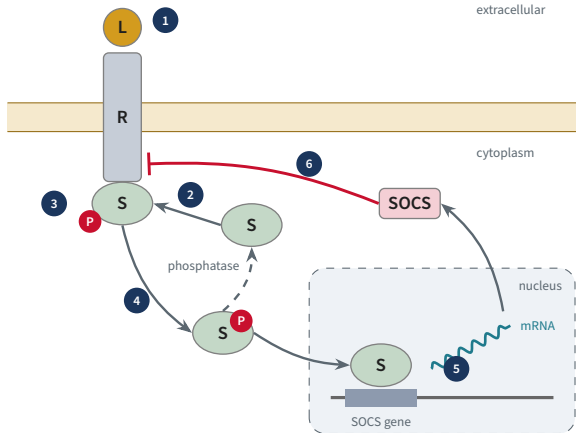


Part 2

A cytokine signaling circuit

Cytokines signal through receptors, STATs, and SOCS feedback

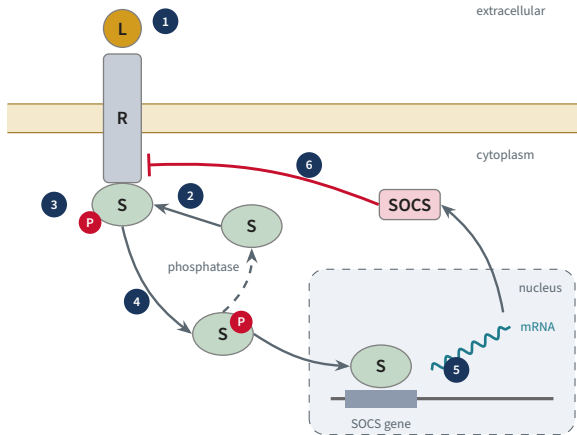
- ▶ **Cytokines** (e.g. interleukins, interferons) are secreted proteins that coordinate immune responses.
- ▶ Binding to their **receptor** activates receptor-associated JAK kinases.
- ▶ **STAT** transcription factors dock on the active receptor, are phosphorylated, and move to the nucleus.
- ▶ pSTAT induces genes including **SOCS** (suppressor of cytokine signaling).
- ▶ SOCS binds the receptor and **shuts off further STAT activation**: negative feedback.



We will build the model one reaction step at a time

1. Ligand-receptor binding
2. STAT binds the ligand-bound receptor
3. Receptor-bound STAT is activated
4. Activated STAT dissociates
5. pSTAT binds the SOCS promoter; SOCS is transcribed and translated
6. SOCS binds the receptor and blocks STAT binding

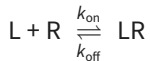
Each step adds reactions to the previous model.
The files are `models/stat_step1_binding.bngl`
through `stat_step6_feedback.bngl`.



Part 3

Step 1: Ligand–receptor binding

Binding is a reversible pair of reactions



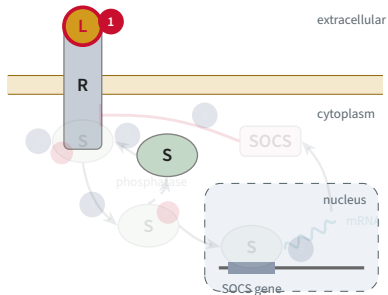
Two elementary reactions, each with its own rate:

binding: $v_1 = k_{on} [L] [R]$

unbinding: $v_2 = k_{\text{off}} [\text{LR}]$

Law of mass action

The rate of an elementary reaction is proportional to the product of the concentrations of its reactants. The proportionality constant is the **rate constant**.



Units: k_{on} in $\mu\text{M}^{-1}\text{min}^{-1}$, k_{off} in min^{-1}

Each reaction contributes to the rate of change of its species

	v_1 (bind)	v_2 (unbind)
L	-1	+1
R	-1	+1
LR	+1	-1

Reactants are consumed (-), products are created (+).

$$\frac{d[L]}{dt} = -k_{\text{on}}[L][R] + k_{\text{off}}[LR]$$

$$\frac{d[R]}{dt} = -k_{\text{on}}[L][R] + k_{\text{off}}[LR]$$

$$\frac{d[LR]}{dt} = +k_{\text{on}}[L][R] - k_{\text{off}}[LR]$$

Ordinary differential equations (ODEs) follow mechanically from the list of reactions and their rate laws.

The stoichiometry matrix gives a compact general form

Collect the concentrations in a vector $\mathbf{X}$ and the reaction rates in $\mathbf{v}$:

$$\frac{d\mathbf{X}}{dt} = \mathbf{S} \mathbf{v}(\mathbf{X}), \quad \frac{d}{dt} \begin{pmatrix} [L] \\ [R] \\ [LR] \end{pmatrix} = \underbrace{\begin{pmatrix} -1 & +1 \\ -1 & +1 \\ +1 & -1 \end{pmatrix}}_{\mathbf{S}} \underbrace{\begin{pmatrix} k_{\text{on}}[L][R] \\ k_{\text{off}}[LR] \end{pmatrix}}_{\mathbf{v}(\mathbf{X})}$$

S_{ij} = (times species i is a product of reaction j) – (times species i is a reactant of reaction j)

- ▶ $\mathbf{S}$ has one row per species and one column per reaction.
- ▶ This form holds for *any* reaction network; only $\mathbf{S}$ and $\mathbf{v}$ change.

Conservation laws and equilibrium

Conservation. Binding only moves molecules between free and bound forms:

$$[L] + [LR] = L_0$$

$$[R] + [LR] = R_0$$

so only one of the three ODEs is independent.

These appear as linear dependencies among the rows of $\mathbf{S}$.

Equilibrium. At steady state $v_1 = v_2$:

$$\frac{[L][R]}{[LR]} = \frac{k_{\text{off}}}{k_{\text{on}}} \equiv K_D$$

Substituting the conservation laws gives a quadratic for $[LR]$:

$$[LR]_{\text{eq}} = \frac{1}{2} \left[b - \sqrt{b^2 - 4L_0R_0} \right],$$

$$b = L_0 + R_0 + K_D$$

Small models can be solved by hand. Real signaling networks cannot, so we solve the ODEs numerically.

ODEs for reaction networks are solved numerically

Simplest idea (forward Euler): take small time steps Δt ,

$$\mathbf{X}(t + \Delta t) \approx \mathbf{X}(t) + \Delta t \mathbf{S} \mathbf{v}(\mathbf{X}(t))$$

In practice:

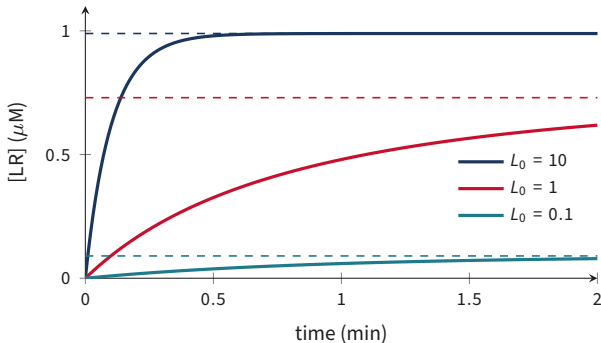
- ▶ The equations are **nonlinear** and usually have no closed-form solution.
- ▶ Reactions occur on very different time scales, which makes the system **stiff**: simple methods need impractically small steps.
- ▶ Production solvers (CVODE, LSODA) adapt the step size and use implicit methods for stiff systems.

In BioNetGen

```
simulate({method=>"ode", ...})  
uses CVODE.
```

In Python, `model.setup_simulator()` returns a `libRoadRunner` simulator for the same model.

Binding approaches equilibrium faster and further at higher ligand



$$R_0 = 1 \mu\text{M}, k_{\text{on}} = 1 \mu\text{M}^{-1}\text{min}^{-1}, \\ k_{\text{off}} = 0.1 \text{ min}^{-1} (K_D = 0.1 \mu\text{M}).$$

Dashed lines: $[LR]_{\text{eq}}$ from the formula on the previous slide.

The relaxation rate is $k_{\text{on}}([L] + [R]) + k_{\text{off}}$ near equilibrium, so higher ligand also means a faster response.

The same model in BNGL

```
begin model

begin parameters
  L0      10    # ligand (uM)
  R0      1     # receptor (uM)
  kon     1     # binding (1/uM/min)
  koff    0.1   # unbinding (1/min)
end parameters

begin molecule types
  L
  R
  LR
end molecule types

begin species
  L    L0
  R    R0
end species
```

```
begin observables
  Molecules Lfree  L
  Molecules Rfree  R
  Molecules LR     LR
end observables

begin reaction rules
  # Ligand binds receptor
  LRbind: L + R <-> LR kon, koff
end reaction rules

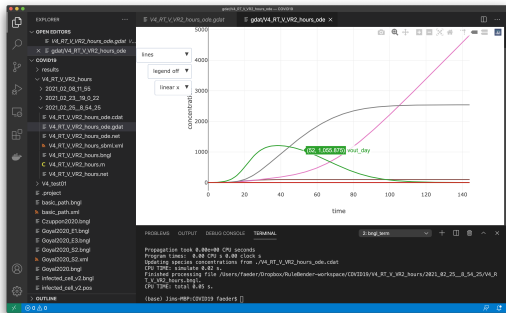
end model

generate_network({overwrite=>1})
simulate({method=>"ode", t_end=>2, n_steps=>200})
```

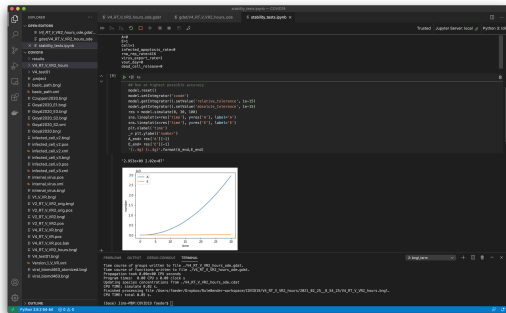
- ▶ **observables:** quantities to output
- ▶ **reaction rules:** $\leftrightarrow$ is reversible; forward and reverse rate constants
- ▶ **actions** after `end model`: generate the network, then simulate



Interactive plotting and Python notebooks



Built-in plotting of .gdat output



Deeper analysis in Jupyter with PyBioNetGen +
libRoadRunner

Part 4

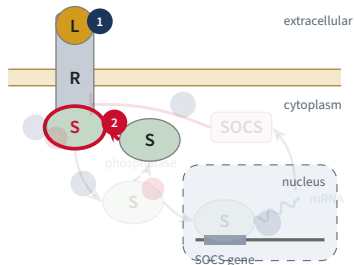
Building the signaling circuit in BNGL

Step 2: STAT binds the ligand-bound receptor



New lines in `stat_step2_recruit.bngl`:

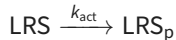
```
begin parameters
  S0      10    # STAT (uM)
  kSon    1     # STAT binding (1/uM/min)
  kSoff   1     # STAT unbinding (1/min)
end parameters
begin molecule types
  S
  LRS
end molecule types
begin species
  S S0
end species
begin reaction rules
  Sbind: LR + S <-> LRS kSon, kSoff
end reaction rules
```



STAT can bind *only* the ligand-bound receptor LR, never free R. This is how ligand binding **enables** STAT recruitment.

Every new species needs a molecule type; species present at $t = 0$ also need an initial amount.

Step 3: receptor-bound STAT is activated

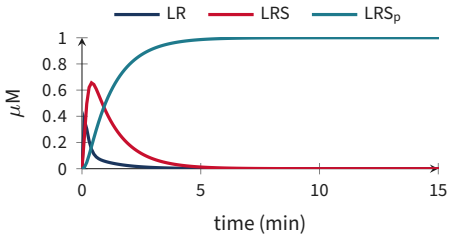
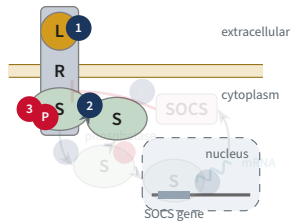


A receptor-associated kinase (JAK) phosphorylates the docked STAT: a **first-order**, irreversible reaction with $v = k_{\text{act}}[\text{LRS}]$.

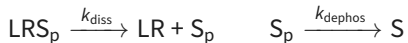
```
kact 1 # STAT activation (1/min)
LRSp
Sact: LRS -> LRSp kact
```

one new line each in parameters, molecule types, and reaction rules

Problem: pSTAT never leaves the receptor. Every receptor ends up stuck as LRS_p .



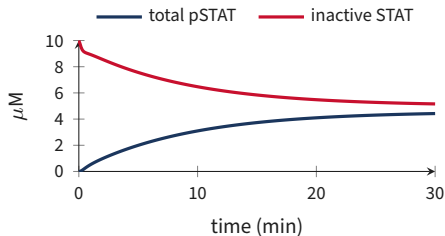
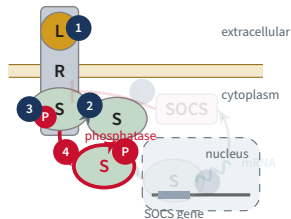
Step 4: activated STAT dissociates from the receptor



Releasing pSTAT frees the receptor to activate another STAT. A phosphatase returns free pSTAT to the inactive pool.

```
kdiss    1    # pSTAT release (1/min)
kdephos  0.1  # dephosphorylation (1/min)
Sp
Srelease: LRSp -> LR + Sp kdiss
Sdephos:  Sp -> S kdephos
```

pSTAT now reaches a steady state set by the balance of activation and dephosphorylation.

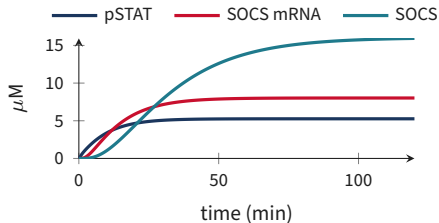
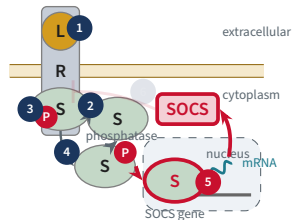


Step 5: pSTAT induces SOCS expression

```
# pSTAT binds the SOCS promoter
Gbind: Sp + G <-> GSp kGon, kGoff
# Bound promoter is transcribed
Transcribe: GSp -> GSp + mSOCS ktx
# mRNA degradation
mRNAdeg: mSOCS -> 0 kmdeg
# Translation
Translate: mSOCS -> mSOCS + SOCS kt1
# SOCS protein degradation
SOCSdeg: SOCS -> 0 kSOCSdeg
```

- **Synthesis:** a species on both sides (GSp, mSOCS) is a template and is not consumed.
- **Degradation:** 0 means nothing; the molecule is removed.

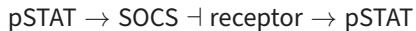
SOCS accumulates, but it has no target yet, so pSTAT is unaffected.



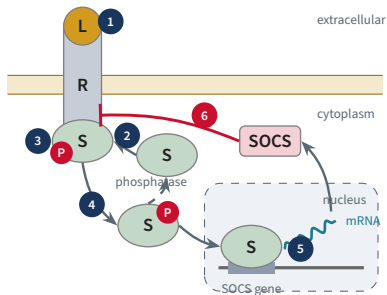
Step 6: SOCS binds the receptor and blocks STAT recruitment



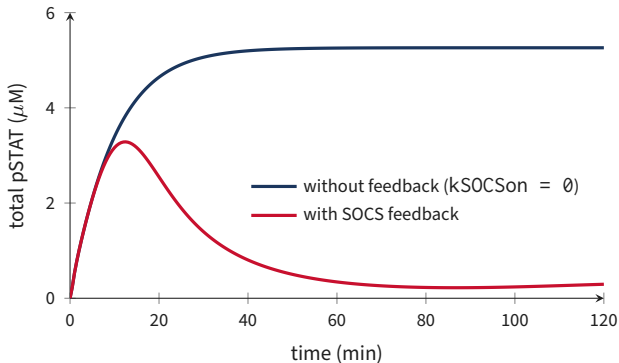
SOCS competes with STAT for the active receptor. LR·SOCS cannot recruit STAT, so pSTAT production drops. The loop is closed:



```
kSOCSon 1 # (1/uM/min)
kSOCSoff 0.01 # (1/min)
LRSOCS
SOCSbind: LR + SOCS <-> LRSOCS kSOCSon, kSOCSoff
```



Negative feedback turns a sustained signal into a pulse



- ▶ Early response is identical: SOCS has not been made yet.
- ▶ As SOCS accumulates it sequesters receptors, and pSTAT falls to a low level despite continued ligand.
- ▶ The delay comes from transcription and translation.

Both curves come from `stat_step6_feedback.bngl`, which runs the model twice using `setParameter`.

The complete network

	reaction	rate
1	$L + R \rightleftharpoons LR$	k_{on}, k_{off}
2	$LR + S \rightleftharpoons LRS$	k_{Son}, k_{Soff}
3	$LRS \rightarrow LRS_p$	k_{act}
4	$LRS_p \rightarrow LR + S_p$	k_{diss}
	$S_p \rightarrow S$	k_{dephos}
5	$S_p + G \rightleftharpoons GS_p$	k_{Gon}, k_{Goff}
	$GS_p \rightarrow GS_p + mSOCS$	k_{tx}
	$mSOCS \rightarrow 0$	k_{mdeg}
	$mSOCS \rightarrow mSOCS + SOCS$	k_{tl}
	$SOCS \rightarrow 0$	$k_{SOCSdeg}$
6	$LR + SOCS \rightleftharpoons LR \cdot SOCS$	$k_{SOCSon}, k_{SOCSoff}$

- ▶ 11 lines of BNGL generate **15 reactions** (4 are reversible) among **12 species**.
- ▶ **S** is a 12×15 matrix; BioNetGen builds it and the ODEs for you (see the .net file).
- ▶ Every species had to be named by hand: L, LR, LRS, LRS_p, LR·SOCS, ...

Part 5

Looking ahead

Naming every complex by hand does not scale

Our model made strong simplifying assumptions. Relax a few of them:

- ▶ SOCS can also bind receptors that already carry STAT (LRS, LRS_p)
- ▶ Ligand can dissociate from *any* receptor complex, not only LR
- ▶ The receptor has two chains, each recruiting its own STAT
- ▶ STATs dimerize before entering the nucleus

Each feature **multiplies** the number of distinct complexes, and each complex needs its own species name and its own copy of every reaction it takes part in.

Module 2: represent molecules with internal structure (binding sites and states) and write rules that generate the network automatically.

ODEs describe average behavior

- ▶ ODEs treat concentrations as continuous and deterministic.
- ▶ In a single cell, some species are present in very small numbers: one or two copies of a gene, a few mRNA molecules.
- ▶ Then random timing of individual reactions matters, and cells with identical parameters respond differently.
- ▶ The **same reaction network** can be simulated stochastically with the Gillespie algorithm:
`simulate({method=>"ssa", ...}).`

We return to stochastic simulation in a later module.

Try it yourself

Using the models in `models/`:

1. Run `stat_step1_binding.bngl` at several values of L_0 and check the equilibrium against the formula for $[LR]_{eq}$.
2. In `stat_step4_release.bngl`, set `kdephos` to 0. What happens to `pSTAT`, and why?
3. In `stat_step6_feedback.bngl`, vary `kt1` (translation rate). How do the peak and the late level of `pSTAT` change?
4. Add a reaction that lets `SOCS` bind `LRS` as well as `LR`. What new species do you need?

Summary

- ▶ A reaction network is defined by its species, reactions, rate laws, and initial conditions.
- ▶ Mass-action rate laws turn the network into ODEs: $d\mathbf{X}/dt = \mathbf{S} \mathbf{v}(\mathbf{X})$.
- ▶ BNGL describes the network directly; BioNetGen generates and integrates the ODEs.
- ▶ Building the cytokine–STAT–SOCS circuit one step at a time shows how each mechanism contributes; negative feedback produces an adaptive pulse of pSTAT.
- ▶ Writing every complex as a separate species quickly becomes unmanageable. This motivates rule-based modeling, the subject of Module 2.